

| Field       | Description                                                                            |
|-------------|----------------------------------------------------------------------------------------|
| Bug ID      | BUG-004                                                                                |
| Bug Title   | Session cookie remains valid after user logout                                         |
| Module      | Logout / Session Management                                                            |
| Environment | Browser: Google Chrome 138.0<br>Operating System: macOS 15.x<br>Application: SauceDemo |
| Severity    | Critical                                                                               |
| Priority    | High                                                                                   |
| Status      | New                                                                                    |
| Reported By | M. Ayyan Irfan                                                                         |
| Date        | 3 July 2026                                                                            |

### Bug Description

The application does not invalidate the user's session cookie after logout. The session cookie (session-username=standard\_user) remains valid and can be reused to access authenticated resources. This indicates improper session management and exposes the application to potential session hijacking attacks.

### Preconditions

- User is logged in using valid credentials (standard\_user / secret\_sauce).
- Browser Developer Tools or an intercepting proxy (e.g., Burp Suite) is available.
- The session cookie is copied before logging out.

### Steps to Reproduce

1. Launch the SauceDemo application.
2. Log in using the following credentials:
  - a. **Username:** standard\_user
  - b. **Password:** secret\_sauce
3. Open the browser's **Developer Tools** and copy the value of the session-username cookie.
4. Click the **Menu (≡)** button.
5. Click **Logout**.

6. Using the browser's **Developer Tools**, replace the current cookie with the previously copied session-username cookie.
7. Refresh the page or navigate to a protected page (e.g., the Product Listing page).
8. Observe the application's behavior.

#### Expected Result

The session cookie should be invalidated immediately after logout. Reusing the previous session cookie should not allow access to authenticated resources, and the user should be redirected to the login page or receive an unauthorized response.

#### Actual Result

The previously copied session cookie remains valid after logout. Reusing the cookie allows access to authenticated resources without requiring the user to log in again.

#### Reproducibility

**Always (5/5)**

#### Evidence

- Screenshot showing the session cookie before logout.
- Screenshot of the reused session cookie after logout.
- Screen recording demonstrating that the user can still access authenticated resources after restoring the previous session cookie.